

A Cost-Simulation Feasibility Study of Budget-Conditioned Foveated Adapter Residency under 24GB GPU Memory

Preliminary short-paper draft for research discussion

Abstract

Physical-AI systems often require perception, planning, verification, and world-model components to share a limited GPU memory budget. This note evaluates whether a budget-conditioned foveated adapter residency policy is worth pursuing before implementing full VLM training or HydraLoRA-style fine-tuning. Across 4,032 simulated configurations, the foveated input reduced visual tokens by 75.0% in the primary case, while the final budgeted variant reduced estimated peak memory by 21.9% relative to full-resolution input. The shared-bank and budgeted variants reduced resident adapter memory by 67.36% and 89.24%, respectively, compared with an independent bank.

Keywords: foveated inference; adapter residency; LoRA; GPU memory; cost simulation; physical AI

1. Introduction

A deployment-oriented physical-AI loop is not only a question of running the largest possible vision model. The perception model must leave memory for planning, verification, orchestration, and possibly a world model. This motivates a policy view: the system should decide where to look, which lightweight adapter to activate, and when to keep or evict that adapter under a memory budget.

The proposed direction combines foveated input selection, LoRA-style local skill adapters, and HydraLoRA-style sharing. The central question is modest: before expensive model training, does this structure produce a measurable memory-cost advantage in simulation?

Contributions

- A cost model separates visual-token cost, adapter resident memory, temporary load buffers, and orchestration reserve.
- The study compares full-resolution inference against foveated ROI inference and independent adapter banks against shared adapter banks.
- A stress grid shows that budget-conditioned residency can remove reserve failures in this simulator.

2. Method

The simulator uses a proxy 7B VLM memory profile on an RTX 3090 24GB GPU. The primary case assumes a 12GB base model, a 4GB orchestration reserve target, a 1344 x 1344 full-resolution input, a 336 x 336 global view, three 336 x 336 ROI crops, sixteen candidate adapters, and top-k=2 active adapters.

Six baselines are evaluated: B0 low-resolution global view, B1 full-resolution input, B2 foveated ROI without adapters, B4-lite independent adapter residency, B5-lite shared adapter residency, and B7-lite budget-conditioned residency with top-k, ROI, or adapter-hold actions.

The evaluation grid spans model profiles, reserve targets, input resolutions, ROI counts, adapter counts, and active top-k values. Because this is a cost simulation, no task-accuracy claim is made.

3. Results

In the primary case, B1 full-resolution input used 9216 visual tokens and 17.30GB estimated peak memory. B2 foveated ROI input used 2304 tokens, a 75.0% token reduction. B7-lite reached 13.51GB estimated peak memory, a 21.9% reduction relative to B1.

Baseline	Tokens	Peak GB	Adapter GB	Reserve	Latency
B0	576	12.07	0.00	pass	48.3
B1	9216	17.30	0.00	pass	143.4
B2	2304	13.11	0.00	pass	67.3
B4-lite	2304	16.02	2.81	pass	103.3
B5-lite	2304	14.12	0.92	pass	83.3
B7-lite	2304	13.51	0.30	pass	83.3

Table 1. Primary-case simulation results.

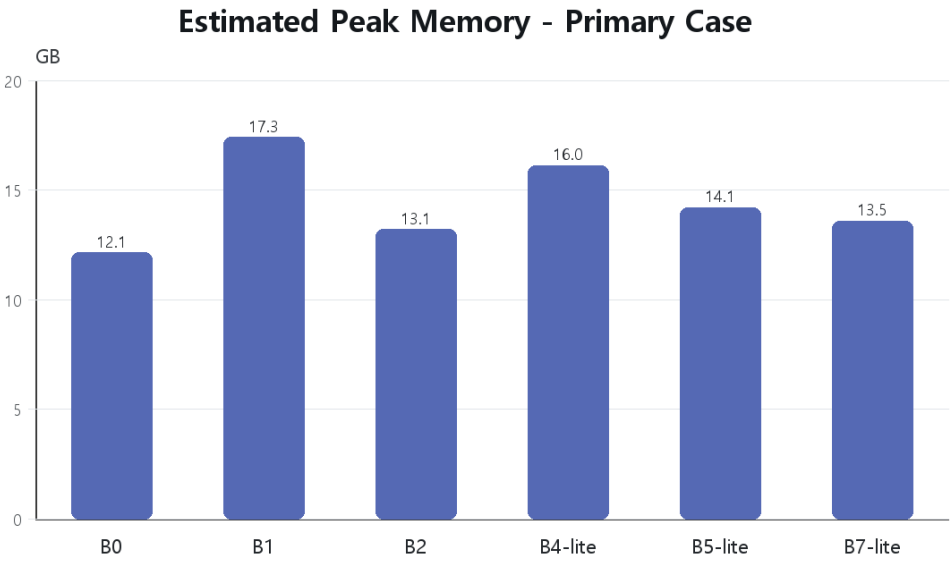


Figure 1. Estimated peak memory by baseline in the primary case.

In the full grid, B4-lite produced 396 reserve failures, whereas B7-lite produced 0 reserve failures and adjusted 148 of 1296 runs through top-k, ROI, or adapter-hold decisions.

4. Discussion

The result supports the feasibility of separating two resource-control levers: foveated input selection reduces visual-token and activation-related cost, while shared or budgeted adapter residency reduces memory tied to skill specialization. A policy formulation is useful because it can trade adapter breadth or ROI count for reserve safety.

Limitations

The simulator is not a profiler. It does not measure kernel-level memory allocation, real VLM attention behavior, adapter loading overhead, or task accuracy. The adapter costs are first-pass assumptions and must be replaced with measured values once a target VLM and LoRA rank are selected.

Conclusion and Next Step

The 24GB GPU cost simulation indicates that budget-conditioned foveated adapter residency is worth validating with real model runs. The immediate next step is to profile a small VLM under full-resolution and foveated inputs, then replace the simulated adapter table with measured LoRA loading and residency costs.

References

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